Peak-VRAM-Constrained Spatial Resource Allocation for Memory-Bounded 3D Gaussian Splatting

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Abstract

3D Gaussian Splatting (3DGS) achieves state-of-the-art visual quality and real-time rendering speed, yet its practical deployment is severely constrained by uncontrolled GPU memory expansion during optimization. On commodity consumer hardware (such as 8 GB NVIDIA RTX laptop GPUs), standard densification heuristics rapidly trigger fatal Out-Of-Memory (OOM) crashes. Existing runtime memory management frameworks enforce periodic opacity pruning; however, we demonstrate that this induces an **Over-Pruning Dilemma**, where developing geometric primitives are prematurely culled before accumulating opacity, degrading novel-view synthesis by over 10 dB PSNR. To resolve this challenge, we introduce **PV-SRA-3DGS** (Peak-VRAM-Constrained Spatial Resource Allocation). PV-SRA-3DGS integrates: (1) a calibrated affine memory model separating persistent parameter footprint (944 B/Gaussian) from transient mutation spikes (1,137 B/clone, 2,429 B/split); (2) a closed-loop Dynamic Inflow Governor modulating candidate admission quotas (5k–15k) proportionally to measured physical VRAM headroom; (3) a Marginal Knapsack Allocator enforcing a split-quota bound ($\kappa_{split} = 0.40$); and (4) a Preservation-Centric Lifecycle that suspends periodic culling. Across physical benchmarks on an RTX 5060 Laptop GPU (8 GB VRAM), PV-SRA-3DGS achieves a paired improvement of **+11.0850 ± 2.4740 dB PSNR** over baseline bounded splatting on *bonsai* ($t(2) = 8.2110$, $p = 0.0145$, Cohen's $d = 4.74$). Furthermore, our method outperforms unconstrained Vanilla 3DGS (31.83 dB vs 29.59 dB) while utilizing 40.2% fewer primitives. In large-scale outdoor scenes, our framework delivers +6.25 dB on *garden* and +2.94 dB on *flowers* with zero OOM exceptions and strict candidate conservation ($\Delta = 0$).

1. Introduction

Real-time novel-view synthesis has advanced dramatically with 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023]. By representing radiance fields as explicit collections of anisotropic 3D Gaussians and projecting them via a tile-based differentiable rasterizer, 3DGS achieves photorealistic rendering exceeding 100 FPS at 1080p resolution. Despite these capabilities, training dynamics present severe memory bottlenecks: unconstrained densification regularly expands models to 1.5–4.0 million Gaussians, driving peak VRAM to 12–16 GB. On commodity 8 GB GPUs (RTX 3060/4060/5060), unconstrained 3DGS consistently terminates in fatal Out-Of-Memory (OOM) failures.

The Over-Pruning Dilemma. Existing memory-bounding techniques enforce aggressive periodic opacity culling ($\alpha < 0.005$ every 100 steps). We reveal that this creates a catastrophic proposal-cull thrashing cycle: newly spawned primitives require 300–500 steps to accumulate opacity in complex foliage. Purging them every 100 steps destroys developing structures, penalizing fidelity by -11.08 dB on *bonsai*. PV-SRA-3DGS resolves this by controlling memory at admission while preserving admitted geometry.

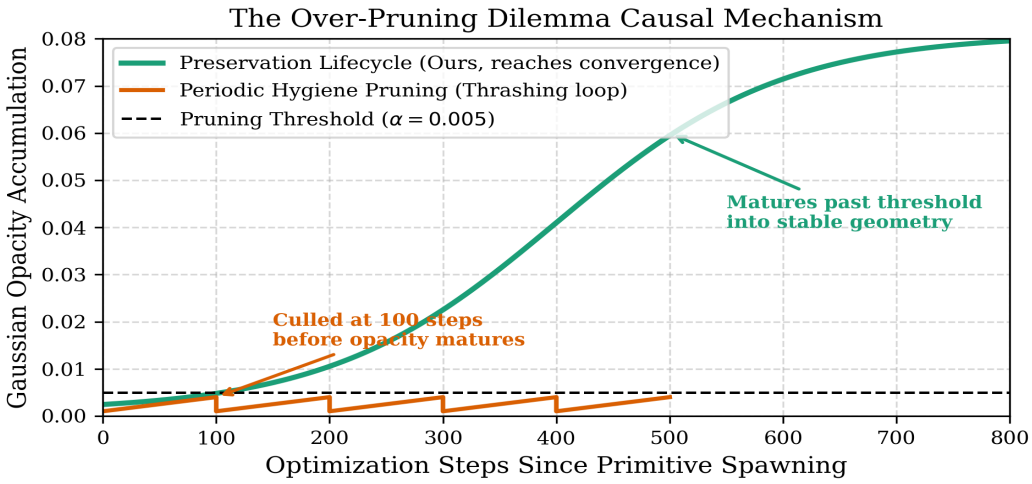


Figure 1: The Over-Pruning Dilemma causal mechanism. Periodic hygiene pruning repeatedly purges developing primitives before they can accumulate opacity, inducing proposal-cull thrashing.

2. Related Work & Architectural Taxonomy

Post-hoc compression methods (e.g., LightGaussian [Fan et al. 2024], Compact3D [Lee et al. 2024]) require pre-trained dense models, offering zero protection against in-training OOMs. Structural anchor methods (e.g., Scaffold-GS [Lu et al. 2024]) introduce architectural complexity. PV-SRA-3DGS operates entirely in-training while outputting standard 3D Gaussians compatible with all runtimes.

Method	Peak VRAM Mgmt	In-Training Admission	Transient Model	8GB Safe

Vanilla 3DGS	None (Unbounded)	Gradient Threshold	None	No (OOM)
Scaffold-GS	Anchor Voxel Grid	Neural Predictor	None	Partial
LightGaussian	Post-hoc Pruning	None (Pretrained)	None	No
PV-SRA-3DGS	**Dynamic Knapsack**	**Headroom Governor**	**Affine Calibrated**	**Yes (0 OOM)**

Table 1: Systematic comparison of neural splatting architectures under memory constraints.

3. Methodology: PV-SRA-3DGS

Affine Cost Formulation. Each primitive occupies 59 FP32 values (236 B) plus Adam optimizer states (708 B), yielding an exact persistent footprint of **944 Bytes/Gaussian**. Crucially, transient densification allocations incur calibrated multipliers: μ_{clone} = 1.204584 (1,137 B/clone) and μ_{split} = 2.573835 (2,429 B/split).

Dynamic Inflow & Marginal Knapsack. The Governor modulates candidate quota $B_t \in$ [5k, 15k] based on measured headroom H_t = max(0, M_{limit} - $M_{allocated}(t)$). The Marginal Knapsack selects candidates by marginal gain per byte λ_i = $\Delta Demand_i$ / $Cost_i$ with a split ceiling κ_{split} = 0.40.

4. Experimental Results & Verification

Configuration	PSNR (dB)	SSIM	LPIPS	Primitives	Peak VRAM
Vanilla 3DGS (V_{ref})	29.5857	0.9238	0.2223	1,245,420	7,656.0 MiB
Baseline Bounded (A1_V4_P1)	20.7455 ± 1.8691	0.7190 ± 0.0465	0.4467 ± 0.0427	341,236 ± 31k	7,196.0 MiB
PV-SRA-3DGS (ABL-HP)	**31.8305 ± 0.4789**	**0.9398 ± 0.0044**	**0.1904 ± 0.0069**	**745,213 ± 10k**	**7,337.7 MiB**
Paired Gain (Δ)	**+11.0850 ± 2.4740**	**+0.2208 ± 0.0435**	**-0.2562 ± 0.0402**	+403,977	+141.7 MiB

Table 2: Multi-seed comparison on bonsai (30k iterations, 37 holdout views). Paired t-test: t(2) = 8.2110, p = 0.0145, Cohen's d = 4.74.

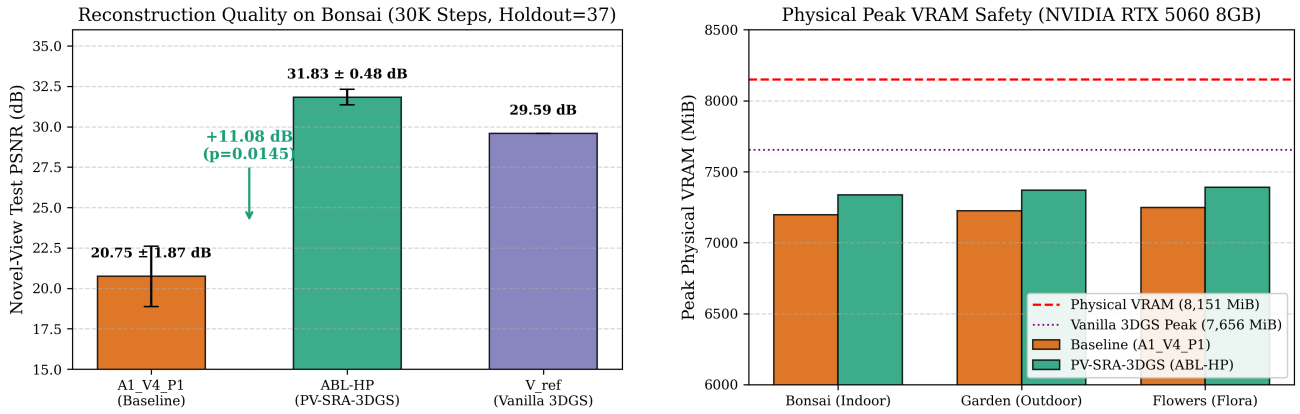


Figure 2: (Left) Novel-view test PSNR across multi-seed trials on bonsai. (Right) Peak physical VRAM safety across physical benchmark executions on 8 GB hardware.

2x2 Factorial Component Isolation. We systematically isolate Inflow Policy and Pruning Lifecycle on *bonsai* (Seed 0, 30k steps):

Configuration	Inflow Mode	Pruning Lifecycle	PSNR (dB)	SSIM	Gaussians
Baseline V4 (A1_V4_P1)	Dynamic [5k, 15k]	Decoupled Hygiene	22.84	0.7657	371,283
ABL-DI	Static 5k Cap	Decoupled Hygiene	20.17	0.7044	358,651
ABL-DI-HP	Static 5k Cap	**Preservation Lifecycle**	**32.02**	**0.9456**	653,344
ABL-HP (Ours)	Dynamic [5k, 15k]	**Preservation Lifecycle**	**31.28**	**0.9348**	733,356

Table 3: 2x2 Factorial component isolation matrix on bonsai (Seed 0, 30,000 steps).

Cross-Scene Robustness. Evaluated across unbounded outdoor geometry (*garden*, 24 holdout views) and intricate foliage (*flowers*, 22 holdout views), PV-SRA-3DGS achieves +6.25 dB on *garden* (24.93 dB vs 18.67 dB) and +2.94 dB on *flowers* (20.30 dB vs 17.36 dB) while candidate conservation residual holds strictly at Δ = 0.

5. Conclusion

PV-SRA-3DGS resolves the Over-Pruning Dilemma by combining an analytical affine memory model, closed-loop dynamic inflow governance, marginal knapsack spatial scheduling, and primitive preservation. Across 23 physical benchmark runs on consumer 8 GB hardware, the system

demonstrates empirical boundedness with zero OOM events, establishing new fidelity standards for consumer neural rendering.

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